Relational Atlas Framework

AMP Note v3: The Odd Preserve-World on the Gauge-Amplitude Trace Basis

Version: v3 (supersedes v2; adds AMP-08: the BCJ / momentum-kernel layer)
Date: June 2026

Naming Convention

This document uses **Relational Atlas Framework (RAF)** for the mathematical and combinatorial layer of the work, and **Emergent Horizon Framework (EHF)** for the broader interpretive hypothesis. This is primarily a RAF document. Its exact claims do not require EHF to be physically correct.

The note references standard objects from tree-level Yang-Mills amplitudes: the trace basis of colour orderings, Parke-Taylor (PT) functions, Kleiss-Kuijf (KK) relations, BCJ relations, the bi-adjoint scalar double-partial amplitudes $m(\alpha|\beta)$ (the inverse of the field-theory KLT momentum kernel), and collinear limits. No physical interpretation of RAF structures is assumed.

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1. Status

This note records the AMP probe series (AMP-01 ... AMP-08), testing whether structures discovered inside the odd preserve-world (OPW) have images in the combinatorics of tree-level gauge amplitudes.

Motivation: the number of canonical Hamilton cycles on K_n, $(n-1)!/2$, is exactly the number of colour orderings of an n-point gauge amplitude modulo cyclicity and reflection. The RAF state space **is** the trace basis modulo reflection; for odd n, partial amplitudes are reflection-odd sections over it.

Headline results, in order of strength:

1. **Sibling-equality theorem with mechanism (AMP-06/07).** Along the exact $7\to9$ core-growth ancestry, the two child chambers of every parent carry equal chamber-local amplitude-relation counts (all 19 pairs). Lemma 1 (proved): the adjacent transposition $(8\,9)$ injects the smaller sibling into the larger with explicit 36-state complement. Lemma 2 (PROVED: single-collinear residue argument plus exact determinant certificates at $n=9$ and $n=11$): chamber relations vanish on that complement. Together: $r_A=r_B$ is an unconditional theorem, and the affine ladder gap 36 is re-derived.

2. **BCJ-level chamber law (AMP-08, new).** The kinematics-dependent BCJ-reduced amplitude span is non-generically aligned with the chambers: chamber-restricted ranks show exact, kinematics-independent deficits (matched random subsets: zero deficit in 100/100 draws). At $n=7$, the BCJ-level chamber-local relation count is **uniform within size classes** — the constant-coefficient irregularities (the 4/4/1 split among 24-chambers) equalise to 4/4/4 — and for sizes ≥24 follows $rb_C = |C| - 20$ exactly.
3. **Confirmed pre-registered prediction (AMP-04/05).** The stabiliser of the chamber-graded relation space at $n=7$ is the boundary group $\langle(2,3),(6,7),\sigma_R\rangle\cong D_4$; the registered forecast $\langle(2,3),(8,9),\sigma_R\rangle$ at $n=9$ was confirmed (residuals 10^{-12}), nearby-swap controls failing at residual ≈0.95.
4. **Fiber-connectivity theorem candidate at $n=7,9,11$.** Chambers coincide exactly with Π-fibers (19/19, 69/69, 251/251); the $n=11$ run independently replicated the documented component data.
5. **Explicit chamber-local amplitude identities (AMP-02/03).** 38 chamber-local KK-type relations at $n=7$; all-±1 basis in the 36-chamber with $\mathbb{Z}_3$ *orbit-difference structure; one identity verified in exact rational arithmetic.*
6. ***($n=9$) relation spectrum (AMP-05).*** *1962 chamber-local relations; σ_R-pairs equal; only the σ_R-fixed central 144-chamber C carries relations among the five 144s.*
7. ***Clean negatives.*** *No amplitude-smoothness from the OPW filter; no KK block-diagonalisation; representation type does not determine r; no parent-(size,r) recursion.*

Claim boundary: these are exact and statistical correspondences between two combinatorial systems. They are not a derivation of gauge theory from RAF. The framework makes no contact with gauge-group selection, matter content, or chirality; Standard-Model and GUT identifications are out of scope.

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2. Setup

2.1 Trace-basis identification

Cyclic orderings $\sigma=(1,a_1,\dots,a_{n-1})$ *with leg 1 first index the trace basis; $A(\sigma^{\mathrm{rev}})=(-1)^nA(\sigma)$. Identifying reversal pairs gives the $(n-1)!/2$ canonical Hamilton cycles. For odd n, amplitudes are reflection-odd sections over the RAF state space.*

2.2 The two relation layers

Constant-coefficient layer (KK). PT functions span, over generic kinematics, the KK-independent subspace U of dimension $(n-2)!$; its annihilator is generated by reflection and KK relations. Constant-coefficient relations satisfied by all PT_λ *hold for all tree partial amplitudes in any helicity configuration.*

Kinematics-dependent layer (BCJ). At fixed generic momenta the amplitudes satisfy further BCJ relations, reducing the independent set to $(n-3)!$. The reduced span is realised as the column space V of the bi-adjoint double-partial matrix $m(\alpha|\beta)$ — the inverse of the field-theory KLT momentum kernel that squares gauge amplitudes into gravity. $V\subset U$, $\dim V=(n-3)!$.

2.3 Ordered fibers and the flip-complement pairing

For an ordering with leg 1 at position 0, let $\widehat{\Pi}(\sigma)=(E,O)$ be the ordered pair of even/odd-position label sets. All ordered fibers $F(E,O)$ have equal size $((n-1)/2)!^2$ (36, 576, 14400 at $n=7,9,11$). Splitting by the canonical condition $a_1<a_{n-1}$ gives the exact pairing

$$
|C(\Pi)| + |C(\mathrm{flip}\,\Pi)| = \big(\tfrac{n-1}{2}\big)!^{2},
$$

pairing the entire chamber-size histogram and identifying the unique unrealised Π-value at each n as the flip of the largest chamber.

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3. AMP-01: sanity layer and statistical layer

Kinematic-span ranks equal $(n-2)!$ at $n=5,6,7,8$ (gaps 13–15 decades). Π-membership explains 31.5%/22.0% of $\log|PT|$ variance at $n=7/9$ ($z=16/376$ vs size-matched random partitions); packets explain 13.1%/3.6% ($z=6/133$). Mechanistic caveat: Π constrains which brackets occur. Negative: the OPW filter adds no amplitude-smoothness (kept/rejected mean $|\Delta\log|PT||$ ratio 1.016 at $n=9$; both far below random pairs).

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4. Fiber-Connectivity Theorem Candidate

For odd $n\ge7$, chambers are exactly the Π-fibers, every fiber connected. Exact at $n=7$ (19=19), $n=9$ (69=69), $n=11$ (251=251, full 1,814,400-state construction, which also independently reproduced the documented edge count 7,862,400 and complete size histogram).

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5. AMP-02: the constant-coefficient layer on chambers ($n=7$)

1. No block-diagonalisation: $\dim(U\cap W_i)=0$ for every chamber.
2. Methodological record: exactly size-proportional chamber energies of U are vacuous — relabeling invariance forces a constant projector diagonal. Worked example of a symmetry-trivial perfect number.
3. Chamber-local relation counts r_i (beyond reflection; fresh-kinematics residuals 10^{-12}–10^{-13}):

size	36	32	28	28	24	24	24	20	20	20	16	16	16	others
r_i	11	7	4	4	4	4	1	1	1	0	1	0	0	0

$\sum r_i=38$; generic expectation 0.

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6. AMP-03: explicit identities (exact arithmetic)

The 36-chamber's 11-dimensional relation space has an all-±1 basis (six of support 6, five of support 10), extracted and verified entirely in exact rational arithmetic (exact rank 25 over $\mathbb{Q}$; every relation exactly zero at six fresh rational kinematic points). Explicit identity, valid for all helicities:

$$
A(1,2,5,3,6,4,7)+A(1,3,6,4,7,2,5)+A(1,4,7,2,5,3,6)
= A(1,2,7,4,6,3,5)+A(1,3,5,2,7,4,6)+A(1,4,6,3,5,2,7).
$$

Every support-6 relation is an orbit-sum difference under a chamber-preserving double 3-cycle: the chamber's residual $\mathbb{Z}_3$ *symmetry organises its relations*.

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7. AMP-04: representation theory — falsification and the boundary group

Representation type under residual chamber symmetry does **not** determine r_i (mixed isotypic membership everywhere; the $\mathbb{Z}_2$*-chamber's single relation sits in a 14-copy multiplicity space; the 36-chamber has 3 of 11 dimensions isotypically forced). Exactly one nontrivial relabeling permutes the chamber partition (σ_R); orientation flips conjugate Π), so there is no global* S_{n-1}-module. Nevertheless the global 38-dimensional relation space has exact stabiliser

$$
K_7=\langle(2\,3),(6\,7),\sigma_R\rangle\cong D_4
$$

(decomposition $38=10\cdot2\oplus9+5+3\ \text{signs}\oplus1$), whose non-central elements scramble the chamber partition while preserving the chamber-graded relation space.

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8. AMP-05: the $n=9$ spectrum and the confirmed prediction

Pre-registered forecast confirmed: \((2\,3)\), \((8\,9)\), \(\sigma_R\), \((2\,3)(8\,9)\) preserve the 1962-dimensional \(n=9\) relation space (residuals \(2\times10^{-12}\)); controls \((4\,5),(5\,6),(2\,4)\) fail (\(\approx0.95\)). Caveat: containment proven for tested elements, not exhaustively.

Spectrum (robust to reseeding and sample-size increase; min sv gap 10.5 decades): total **1962**; by size: 576: 177; 540: 141; 504: 110, 110; 468: 90, 90, 79; 432: 90, 90, 59, 59, 57; 396: 59, 59, 40, 37, 37; 360: 40, 40, 37, 37, 29, 29, 18; 324: 40, 29, 29, 18, 18, 10, 10; 288: 29, 29, 18, 10, 10, 10, 10, 2; 252: 10, 10, 10, 10, 2, 2, 0; 216: 10, 10, 2, 2, 2, 0, 0; 180: 2, 2, 0, 0, 0; 144: 2, 0, 0, 0, 0; \(\le108\): all 0. \(\sigma_R\)-pairs equal (all 27); only the \(\sigma_R\)-fixed central 144-chamber \(C\) carries relations among the five 144s.

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9. AMP-06: the ladder test

Using the independently reconstructed \(7\to9\) core-growth ancestry (720 core descendants; every parent exactly two children, sizes \(9s+108\), \(9s+144\)):

Sibling-equality law (exact, all 19 pairs). Both children of every parent carry equal \(r\).

Clean negative. Child \(r\) is not a function of parent (size, \(r\)): the two \((28,4)\) parents produce 40 vs 37; \(r{=}0\) parents produce children with \(r\in\{0,2,10\}\). Relations are created by growth; no naive recursion exists.

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10. AMP-07: the sibling mechanism

Lemma 1 (proved). \(\tau=(8\,9)\) maps the canonical states of the smaller sibling \(A\) injectively into the larger sibling \(B\), all signs \(+1\), with complement exactly \(X=\{c\in B: c_1=8,\ c_{n-1}=9\}\), \(|X|=((n-3)/2)!^2=36\). *Proof:* *swapping adjacent values changes no comparison with third values; the canonical condition* \(c_1<c_{n-1}\) can flip only when \(\{c_1,c_{n-1}\}=\{8,9\}\); in \(A\) that configuration is forced anti-canonical; the unhit canonical \(B\)-states are exactly the \(c_1=8,c_{n-1}=9\) pattern, counted by \(3!^2\). \(\square\) Corollary: the ladder gap \(144-108=36\) is \(|X|\) — explained, not observed.

Lemma 2 (PROVED at \(n=9\) and \(n=11\); see §11D). Every chamber-local relation of \(B\) vanishes on \(X\). The proof is a single-collinear residue argument (\(\lambda_8\to\lambda_1\), with canonicality forcing the polar set to be exactly \(X\)) plus an exact determinant certificate for the independence of the reduced \((n-1)\)-point family. The earlier double-collinear sketch is superseded.

Theorem (now unconditional at $7\to9$ and $9\to11$). τ injects $R_A\hookrightarrow R_B$ and $R_B\hookrightarrow R_A$; hence $r_A=r_B$. Both inclusions verified at $\sim10^{-12}$ in all pairs; with Lemmas 1 and 2 both proved, the equality is a theorem.

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11. AMP-08: the BCJ / momentum-kernel layer (new in v3)

11.1 Object and validation chain

The bi-adjoint double-partial matrix $m(\alpha|\beta)$ was computed from first principles as a signed sum over trivalent trees planar with respect to both orderings ($\prod_I 1/s_I$ with relative-winding signs), at exact momentum-conserving spinor kinematics. Validation:

- **Sign convention pinned by rank collapse:** correct signs give rank $(n-3)!$ = 2 at $n=5$ and 6 at $n=6$; the alternative convention gives 11 at $n=5$. The collapse is the diagnostic.
- $n=7$: rank exactly 24; reflection-odd to 10^{-15}.
- **Cross-pipeline check:** the 38 chamber-local relations from the independent PT pipeline annihilate the kernel's columns at 10^{-16}.

11.2 The discovery: non-generic chamber alignment

Let $V=\mathrm{col}\,m$ (dim 24) and let rb_C be the chamber-local relation count at the BCJ level ($rb_C\ge r_C$ since $V\subset U$). Generic expectation for the restricted rank: $\min(24,\ |C|-r_C)$.

Control: random coordinate subsets of matched sizes show **zero** deficit in 100/100 draws. **Chambers:** exact deficits, identical at three independent kinematic points, sv gaps 13+ decades:

chamber size	36	32	28, 28	24, 24, 24	20, 20, 20	≤16
PT-level r	11	7	4, 4	1, 4, 4	1, 0, 1	as §5
BCJ-level rb	16	12	8, 8	**4, 4, 4**	**1, 1, 1**	$=r$

11.3 The emergent law and its anomaly

Two exact observations at $n=7$:

1. **Equalisation.** rb is uniform within each size class for $|C|\ge20$. The constant-coefficient irregularities that defeated representation theory (the 4/4/1 split among 24-chambers; 1/0/1 among 20-chambers) equalise at the BCJ level to 4/4/4 and 1/1/1. The PT-level irregularities appear to be constant-coefficient shadows of a size-uniform kinematic structure.
2. **The cap.** For $|C|\ge24$: $rb_C = |C| - 20$ exactly — the BCJ span sees precisely 20 dimensions in every large chamber. The size-20 class is the boundary case and carries $rb=1$ where $|C|-20=0$; recorded as an honest anomaly in the clean pattern.

No common vanishing subspace: the dimension of V vanishing jointly on the deficit chambers is 0; each chamber pins a different piece of V, in general position. The alignment is genuinely per-chamber.

11.4 The cap constant — both registered hypotheses falsified (AMP-09 outcome)

Two structurally motivated readings of the $n=7$ constant 20 were registered before any $n=9$ computation: **H1** (fiber-count cap $\binom{8}{4}=70$ at $n=9$) and **H2** ($(n{-}3)!$-relative cap ≈716, predicting no $n=9$ deficits). AMP-09 (per-chamber bi-adjoint at $n=9$; global-rank validation 720 with 11.8-decade gap; PT cross-checks 10^{-13}–10^{-15}; matched random-row control at full rank) falsified **both**:

- H1 died immediately: 144-chambers measure full rank 144, not 70.
- H2 died at the giants: the 576- and 504-chambers cap at measured rank **342**, not 399/394.

The measured law across ten chambers (sizes 144, 144, 144, 288, 396, 432, 432, 504, 540, 576; all gaps ≥11.2 decades):

$$\mathrm{rank}\big(V|_C\big) = \min\big(342,\ |C|-r_C\big), \qquad\text{equivalently}\qquad rb_C = \max\big(r_C,\ |C|-342\big).$$

A further registered sub-prediction also failed: the $n=7$ boundary anomaly (the size-20 chamber's $rb=1$) suggested boundary chambers ($|C|-r_C=342$, the two $r{=}90$ 432-chambers) would measure 341; both measure the clean 342. The $n=7$ size-20 anomaly therefore stands isolated and unexplained.

Open constant. $\mathrm{cap}(7)=20$, $\mathrm{cap}(9)=342$. No formula is claimed; two data points underdetermine it, and the next constraints should come from structural probes (the per-chamber kernels of V, dims 4 and 378; fiber-union restrictions) rather than from numerological fits.

11.5 Significance

This is the framework's first contact with the kinematics-dependent layer of amplitude structure — the layer that the KLT kernel squares into gravity. The chamber decomposition now interacts exactly with both floors: constant-coefficient (KK) relations and momentum-kernel (BCJ) structure.

11A. AMP-11: the fiber law (new)

11A.1 Covariance theorem for fiber-unions

Individual chambers are not relabeling-covariant (\u00a77.2), but the **flip-pairs** \(\{C(\Pi), C(\mathrm{flip}\,\Pi)\}\) are: a relabeling \(g\) (fixing label 1) commutes with \(\mathrm{flip}\), so \(g\) maps the cycle-level fiber-union \(C\cup C_{\mathrm{flip}}\) of \(\Pi\) to that of \(g\Pi\). *Consequently every relabeling-covariant rank invariant is **forced uniform** across all fiber values (\(\binom{6}{3}=20\) at \(n=7\); \(\binom{8}{4}=70\) at \(n=9\)) — exactly at the constant-coefficient level, and at generic kinematics for the BCJ level.*

11A.2 The PT fiber law and cross-chamber relations

Measured: \(\mathrm{rank}(U|_{\text{fiber}})\) *equals **25** for all 10 fiber pairs at \(n=7\) (gaps 15.1 decades) and **399** for the three tested pairs at \(n=9\) (gaps 14.4). The fiber relation count*

$$\kappa(n) = m!^2 - \mathrm{rank}(U|_{\text{fiber}}),\qquad \kappa(7)=11,\quad \kappa(9)=177,$$

is the uniform invariant, and \(\kappa(n) = r(\text{largest chamber})\) since the largest chamber's flip partner is the empty fiber. For every other pair, the local counts fall short and the deficit is carried by **cross-chamber relations** binding the two complementary chambers: \(\mathrm{CROSS} = \kappa - r_C - r_{C_{\mathrm{flip}}}\) (at \(n=7\): 0, 4, 7, 7, 7, 10, 10, 10, 10 across the pairs; at \(n=9\): 0, 36, 67 on the tested pairs). The chamber-local relation spaces of AMP-02/05 are therefore the *diagonal part* of a uniform fiber-level structure.

11A.3 The BCJ fiber law: the cap explained structurally

Measured at \(n=7\): every flip-pair union has BCJ rank exactly **20** (= the cap), and the kernels \(K_C=\{v\in V: v|_C=0\}\) *of all large chambers coincide with the 4-dimensional **fiber kernel** (e.g. the (32,4) pair: \(\dim(K_{32}\cap K_4)=4=\dim K_{32}\)). Confirmed at \(n=9\): the unions \((540,36)\) and \((504,72)\) both measure BCJ rank exactly **342** (gaps 11.4/11.6 decades) against generic values 378/414. Hence*

$$\mathrm{cap}(n) = \mathrm{rank}(V|_{\text{fiber}}),$$

uniform by covariance, and the chamber cap law \(\mathrm{rank}(V|_C)=\min(\mathrm{cap},|C|-r_C)\) *follows from \(C\subset\) fiber-union.*

11A.4 The constants ledger and the route to formulas

invariant	\(n=7\) (\(m=3\))	\(n=9\) (\(m=4\))
fiber size \(m!^2\)	36	576
PT fiber rank	25	399
\(\kappa\) (PT fiber relations)	11	177
BCJ fiber rank (= cap)	20	342
BCJ fiber kernel in \(V\)	4	378

No formulas are claimed. The proper route: the fiber-union carries the residual symmetry of \u00a77.3-type groups, and the fiber is an \(S_m\times S_m\)-torsor, so all four invariants are dimensions of explicit \(S_m\times S_m\)-bimodule constructions — a finite, well-posed representation-theoretic computation, not a curve fit. (Recorded observation, no weight attached: \(25=5^2\) and \(399=20^2-1\).)

11B. AMP-12: the fiber relation spaces as \(S_m\times S_m\) modules (new)

11B.1 Setup and validation

The fiber-union stabiliser contains \(S_{E'}\times S_O\cong S_m\times S_m\) *(relabelings within the two alternating sets). The fiber relation spaces were decomposed under this group via the signed cycle action: equivariance errors \(\sim10^{-15}\); the multiplicity tables are identical across all 10 fiber pairs at \(n=7\) (as the covariance theorem requires); dimension checks exact (11 and 177).*

11B.2 The multiplicity tables

The FULL signed fiber module is the **regular representation**: multiplicities exactly \(d_\lambda d_\mu\) (verified at both \(n\)). The relation-side multiplicities \(m_R(\lambda,\mu)\) (span side \(=d_\lambda d_\mu-m_R\)):

\(n=7\), rows \(\lambda\), columns \(\mu\) over \((3),(21),(1^3)\):

	(3)	(21)	(1³)
(3)	0	0	1
(21)	0	1	1
(1³)	1	1	1

Recorded observation: support \(=\{\mu\trianglelefteq\lambda'\}\) (dominance below the conjugate), all multiplicities 1.

\(n=9\), over \((4),(31),(22),(211),(1^4)\):

	(4)	(31)	(22)	(211)	(1⁴)
(4)	0	0	0	1	1
(31)	0	2	0	3	2
(22)	0	0	1	2	2
(211)	1	3	2	4	2
(1⁴)	1	2	2	2	1

Symmetric, as the swap symmetry requires. The \(n=7\) dominance rule does **not** persist: e.g. \(m((4),(211))=1\) although \((211)\not\trianglelefteq(1111)\). No closed form is claimed.

11B.3 The KK span is the Lie module

The character of \(U\) under relabelings of legs \(2,\dots,n\) matches the classical Lie-module character \(\chi_{\mathrm{Lie}(n-1)}\) *exactly on all 11 conjugacy classes at \(n=7\) (values \(120,-8,-3,+1\) on the uniform cycle types \(1^6,2^3,3^2,6\); zero elsewhere; machine precision). This is Ree's theorem in amplitude language (KK relations are shuffle relations; words modulo shuffles form the free Lie coalgebra) and is presumably known via the DDM construction; it is recorded here as a verified structural anchor, not a novelty claim.*

11B.4 Multiplicity data is provably insufficient

With \(U\cong\mathrm{Lie}(n-1)\), the restriction multiplicities \(m_{\mathrm{Res}\,U}(\lambda,\mu)\) *are analytic (Brandt character); at \(n=9\) they range from 8 to 78 and exceed \(d_\lambda d_\mu\) on every block. A blockwise-generic equivariant evaluation map \(W\to U^*\) would therefore have full block ranks, forcing \(m_{\mathrm{span}}=d_\lambda d_\mu\) and \(\kappa=0\). The observed \(\kappa=11,177\) hence* ****cannot be a function of the multiplicity data of the modules involved****: *the fiber relation content is a property of the specific evaluation map (the Parke–Taylor pairing), not of the representation types. This is the module-level analogue of the chamber-level negative of §7.1.*

11B.5 Ruled out, and avenues

Ruled out as closed forms for \(m_R\): Kostka numbers (wrong values at \(n=7\)); restrictions of fixed \(S_{n-1}\)*-irreducibles (no dimension match at 177; extensions are not determined by restriction data — the available class equations are underdetermined); pure Kronecker forms (total dimension divisible by \(m!\), but \(24\nmid177\)). Open avenues: refinement under the swap-extended group \((S_m\times S_m)\rtimes\mathbb{Z}_2\)*; analysis of the evaluation map in Specht bases; explicit exact extraction of cross-chamber relations (AMP-03 methodology applied to a fiber pair with CROSS \(>0\)); a \(\kappa(11)\) prediction to be registered only if a rule emerges.

11C. AMP-13: explicit cross-chamber identities (new)

11C.1 Result

On the \((32,4)\) fiber pair at \(n=7\) (fiber \(E,O=\{1,4,6,7\}\,|\,\{2,3,5\}\)); CROSS \(=\kappa-r_{32}-r_4=4\)*), the entire computation was carried out in exact rational arithmetic (AMP-03 methodology). Findings, all exact over \(\mathbb{Q}\):*

1. The fiber-union PT matrix has exact rank **25** over \(\mathbb{Q}\) — the fiber law upgraded from float to exact for this pair; relation lattice dimension 11.

2. The relation lattice projects **onto** the four tiny-chamber coordinates with **unit pivots**. Consequently each tiny-chamber amplitude is an integer-coefficient — in fact all-±1 — combination of 32-chamber amplitudes, valid for all tree-level partial amplitudes in any helicity configuration.
3. The four canonical cross relations, reduced against the 7-dimensional local lattice, have supports 6, 6, 6, 10 with all coefficients ±1. Explicit example:

$$
A(1,4,2,6,3,7,5) = A(1,2,4,5,7,3,6) - A(1,2,6,3,7,5,4) + A(1,3,6,2,4,5,7) - A(1,3,7,5,4,2,6) - A(1,5,4,2,6,3,7).
$$

All six orderings lie in the same fiber pair; the left side is a state of the size-4 chamber, the right side entirely in the complementary 32-chamber.

4. All 11 relations (4 cross + 7 local) evaluate to the exact rational number 0 at five fresh random rational kinematic points.
5. The four cross relations form a single orbit family under the pair stabiliser $S_{E'}\times S_O$ *(verified exactly on a generator mapping tiny state 0 to tiny state 1).*

11C.2 Remarks

The onto-projection is special to pairs with CROSS $=|C_{\mathrm{flip}}|$ *(here $4=4$). For the $(28,8)$ pair, CROSS $=7<8$*: the projection has a one-dimensional cokernel, i.e. exactly one combination of the 8-chamber amplitudes is *not* expressible from the 28-chamber — a distinguished covector on the small chamber whose identification is left open.

These identities are constant-coefficient and therefore lie in the Kleiss-Kuijf span; their content is the fiber-pair support structure, which is the RAF-derived grading. Literature placement before any novelty claim, as throughout.

11C.3 AMP-14: the distinguished covector of the (28,8) pair

*The cokernel dimensions $|C_{\mathrm{flip}}|-\mathrm{CROSS}$ across the $n=7$ pairs are: $(36,0)\!:0$; $(32,4)\!:0$; $(28,8)\!:1,1$; $(24,12)\!:2,5,5$; $(20,16)\!:6,6,6$. The two $(28,8)$ pairs (σ_R partners) are the unique one-dimensional cases, so each carries a **canonical covector** φ — the unique (up to scale) element of the kinematic row space supported entirely on the small chamber. Extracted in exact rational arithmetic (rank 25 over $\mathbb{Q}$) re-confirmed; nullspace dimension exactly 1):*

$$
\varphi = A(1,3,4,6,2,7,5)+A(1,3,4,7,2,6,5)+A(1,3,5,6,2,7,4)+A(1,3,5,7,2,6,4),
$$

i.e. all-$(+1)$ coefficients on exactly the four 8-chamber states with **label 2 at the interior position 4**, and zero on the four with label 2 at position 2. (In this chamber, canonicality forces $c_1=3$), so position 2 vs 4 is the only freedom for the smallest label.) φ annihilates the full 11-dimensional relation lattice exactly, and transforms **trivially** under the 8-chamber's order-4 stabiliser — it is the orbit-indicator of the single support orbit.

Operational meaning: a combination $\sum_x t_x A(x)$ of 8-chamber amplitudes is expressible in 28-chamber amplitudes **iff** $\varphi\cdot t=0$. The single obstruction to cross-chamber expressibility is the smallest-label-interior orbit sum.

BCJ visibility: no. With the correct complex projector, the BCJ evaluation span on the union has rank 20 (fiber law re-confirmed, 12.9-decade gap), and φ lies outside it at generic kinematics (residuals 0.44 and 0.13 at two independent kinematic points). φ is a purely constant-coefficient-layer object. Numerics note, recorded per house rules: an initial check returned "yes" with an apparent rank 25; both numbers were artifacts of stacking real and imaginary parts as columns (which measures the real form of the complex span, not the span); the corrected complex-projector computation reverses the verdict and restores the rank-20 law.

Open: derive φ's combinatorial form from the evaluation-map / collinear structure; the $(24,12)$ pair with cokernel 2 is the next-most-canonical object.

11D. AMP-15: proof of the vanishing lemma, and the S-channel structure (new)

11D.1 Theorem and proof

Theorem (vanishing lemma, $n=9$ and $n=11$). Let B be a B-form chamber ($n\in E_B$, $n-1\in O_B$) and let w, supported on the canonical states of B, satisfy $\sum_c w(c)\,\mathrm{PT}_{n(c)\equiv 0}$ *identically in the spinors. Then* $w|_X=0$, where $X=\{c\in B: c_1=n-1,\ c_{n-1}=n\}$.

Proof. (i) Substitute $\lambda_{n-1}=\lambda_1+\varepsilon\mu$, all other spinors generic. The only vanishing bracket is $\langle 1,n-1\rangle$; every $\langle n-1,j\rangle\to\langle 1,j\rangle\neq 0$, including $j=n$. A state of B has label $n-1$ adjacent to label 1 iff $c_1=n-1$: label $n-1$ occupies odd positions, and the only odd position adjacent to position 0 is position 1. (ii) Canonicality $c_1<c_{n-1}$, with $c_{n-1}\in E_B$ and n the only E_B-label exceeding $n-1$, then forces $c_{n-1}=n$: the polar set is exactly X. (iii) For $x\in X$, $\mathrm{PT}_n(x)=\mathrm{PT}_{n-1}(\hat x)\,(\varepsilon\langle 1\mu\rangle)^{-1}(1+O(\varepsilon))$, where $\hat x$ deletes label $n-1$; the polar factor is uniform over X and $x\mapsto\hat x$ is injective. The ε^{-1} coefficient of the identity gives $\sum_{x\in X}w(x)\,\mathrm{PT}_{n-1}(\hat x)\equiv 0$ identically in the remaining spinors. (iv) The reduced family (36 orderings at $n=9$; 576 at $n=11$; label n fixed at position $n-2$) contains no reversal pairs — a reversal places n at position 1, outside the family — and is linearly independent: certified by a nonvanishing 36×36 determinant of exact PT values at integer spinor kinematics modulo $p=2147483647$ and independently modulo $p=2147483629$ (and a 576×576 certificate at $n=11$). A nonzero determinant mod p implies a nonzero determinant over $\mathbb{Q}$, hence $\mathbb{Q}$-independence, hence $\mathbb{C}$-independence (the family is defined over $\mathbb{Q}$). Therefore $w|_X=0$. $\square$

Relabeling covariance makes one certificate per n cover all sibling pairs at that n. Float cross-check: the 36-family is full-rank with no small singular values whatsoever. The analytic steps (i)–(iii) are uniform in odd n; the all-n statement is thereby reduced to one determinant per n, with a uniform independence proof the remaining open step.

Corollary (sibling equality, unconditional). With Lemma 1, $r_A=r_B$ for all 19 sibling pairs at $7\to 9$ and all $(10,11)$-split pairs at $9\to 11$.

11D.2 The S-channel: a falsified pre-registered prediction and a new even-n constant

Registered before testing: the channel $\lambda_n\to\lambda_1$ has polar set $S=\{c_{n-1}=n\}$ *(all such tuples are canonical, so $|S|=144$ inside every B-form chamber at $n=9$)), with pushforward bijective onto a reversal-closed class of 144 eight-point orderings. Since* PT_8 is reversal-even, the residue constrains only the symmetrised weights over the 72 cycle pairs; **if** the 72 cycle-level functions were independent, every chamber relation would satisfy $w(c)=-w(\mathrm{partner}(c))$ on S (P–S antisymmetry) and, with Lemma 2, vanish on $Y_2=\{c_{n-2}=n-1,\ c_{n-1}=n\}$.

Outcome: falsified, consistently. The 144-class has rank **57**: the 72 cycle-level functions satisfy $\kappa_8^{\mathrm{even}}=15$ relations — the first even-n fiber relation count measured in this programme, a new ledger constant. The mod-p certificates correctly fail, and across all 16 relation-bearing B-form chambers the P-S residual and $|w|$ on Y_2 are order one. The surviving (weaker) law: the symmetrised S-weights of every chamber relation land in the 15-dimensional even-fiber relation space.

B-form sweep at $n=9$ (bonus): the 16 relation-bearing B-form chambers, $(|B|,r)$ = (468,90), (432,59), (396,40), (396,37), (360,40), (360,29), (360,18), (324,18), (324,10), (324,10), (288,10), (288,10), (288,2), (252,10), (252,2), (216,2), all with $\max|w|$ on $X\approx10^{-15}$ — the theorem's content re-confirmed comprehensively.

11E. AMP-16: the n=11 boundary stabiliser — prediction resolved and proved (new)

11E.1 Result

Theorem (boundary stabiliser, $\supseteq$-direction, all of $n=7,9,11$).
$K_n\supseteq\langle(2\,3),(n-1\;n),\sigma_R\rangle$, where K_n is the stabiliser of the chamber-graded relation space. Registered prediction §12.2 ($K_{11}\supseteq\langle(2\,3),(10\,11),\sigma_R\rangle$) *is thereby* ***confirmed and upgraded to a theorem***. *Each generator is established by a mechanism, not a numeric coincidence.*

11E.2 The three mechanisms

(a) $\sigma_R\in K_n$, all n (theorem; discovered here). For any canonical state, $c_1<c_{n-1}$ gives $(n+2)-c_1>(n+2)-c_{n-1}$, *so the reversal relabeling $j\mapsto(n+2)-j$ sends every canonical state to a canonical state (no reversal-to-restore needed). Hence σ_R permutes whole chambers, and by the covariance theorem (§11A.1) it carries each chamber-local relation space to another chamber's relation space bijectively — it stabilises the graded space. No test required; this supersedes the former "registered for $n=11$" status of σ_R.*

(b) $(n-1\;n)\in K_n$ (theorem, via Lemma 2). The top transposition splits only B-form chambers, and the only states it moves out of their chamber are those at the X-corner $\{c_1=n-1,\ c_{n-1}=n\}$ — *exactly the set the vanishing lemma (Lemma 2, §11D) annihilates. So every relation maps to a union relation whose single moved block has zero weight: a clean chamber-respecting image. This retroactively explains $(8\,9)\in K_9$ (AMP-04/05) as the same mechanism.* ∗*Witness:*∗ *on the relation-bearing B-form chamber B (4032 states, fiber from the AMP-10 pair), $(10\,11)$ produces a 576-state block in the 10944-chamber at mass 10^{-14} (NULL) and a 3456-state main block verified as a relation at residual 1.3×10^{-15}.*

(c) $(2,3)\in K_n$ (theorem, via Lemma 2$^\prime$). This is the generator with genuine content, since (a) and (b) reduce to known/structural facts. $(2,3)$ splits chambers with $2\in O,\ 3\in E$, moving out exactly the small-label corner $\{c_1=2,\ c_{n-1}=3\}$.

Decisive numerical witness (C605). The relation-bearing 3456-chamber (fiber 591, $r=6$) was transported by the non-splitting relabeling $(2,5)$ to chamber C605 (5184 states, $2\in O,3\in E$), $r=6$). Applying $(2,3)$: the 576-state corner block lands in the 9792-chamber at mass 8.4×10^{-13} (NULL); the surviving 4608-state block is a genuine relation of the 4608-chamber at residual 6.9×10^{-14}. Controls $(5,7),(6,7),(3,10)$ — all chosen to split — produce $O(1)$ corner mass and components that are **not** relations (residuals 0.1–0.3, at the random-vector baseline), confirming the discrimination is real.

Comprehensive sweep (n=9). The small-label vanishing law holds on **16/16** relation-bearing B-form chambers with $2\in O,3\in E$: in every chamber, from $(468,90)$ down to $(216,2)$, the 36-state corner carries relation mass $\le 1.5\times10^{-15}$.

11E.3 Lemma 2$^\prime$ (the small-label vanishing lemma)

Theorem (Lemma 2$^\prime$, $n=9$ and $n=11$). Let B be a chamber with $2\in O$, $3\in E$, and let w (supported on canonical states of B) satisfy $\sum_c w(c)\,\mathrm{PT}_{n(c)}\equiv0$. *Then w vanishes on the corner $Z=\{c_1=2,\ c_{n-1}=3\}$.*

Proof. The mirror of Lemma 2. Substitute $\lambda_3=\lambda_1+\varepsilon\mu$; the only vanishing bracket is $\langle1,3\rangle$, and label 3 (an even-class label, at even positions) is adjacent to label 1 (position 0) iff $c_{n-1}=3$. *Canonicality* $c_1<c_{n-1}=3$ with $c_1\in O$ forces $c_1=2$: the polar set is exactly Z. The ε^{-1} residue gives $\sum_{z\in Z}w(z)\,\mathrm{PT}_{n-1}(\hat z)\equiv0$, where $\hat z$ deletes label 3 and the map is injective. The reduced family — orderings $(1,2,\,e,o,e,o,\dots)$ with label 2 fixed at position 1, $E\setminus\{1,3\}$ at even positions, $O\setminus\{2\}$ at odd positions ≥3 (36 at $n=9$, 576 at $n=11$) — contains **no reversal pairs** (a reversal moves label 2 off position 1, out of the family) and is independent: certified by a nonvanishing determinant of exact integer-kinematics PT values mod $p=2147483647$ and mod $p=2147483629$ at $n=9$, and mod 2147483647 at $n=11$. Hence $w|_Z=0$. $\square$

The analytic steps are uniform in odd n; the all-n statement reduces (as for Lemma 2) to one determinant certificate per n.

11E.4 Method and numerics lessons

The per-vector test (why no image-side SVD is needed). By covariance, $\rho(g)w$ automatically lies in the *image union's* relation space for any relabeling g; the entire content of "$g\in K$)" is whether each chamber **component** of $\rho(g)w$ is itself a relation, testable by direct PT-annihilation of that component. **Vacuity insight:** if g maps a chamber into a single chamber (no split), the image is automatically a relation and the test is vacuous — so only *splitting* elements discriminate, and a control must be chosen to actually split (the first control battery was inadvertently non-splitting and uninformative).

Gram-eigh failure and the two-stage fix. A single-stage Gram-eigenvalue nullspace on the squared spectrum undercounted the C605 rank (reported $r=5$) vs truth 6) with a self-diagnosing 0.2-decade gap — the squared singular values fell below the `eigh` floor. Fix (validated against the $r=6$) ground truth at 11.4 decades): a two-stage solver — Gram shortlist of ≥25) candidate directions at threshold 10^{-8}), then 400 fresh-kinematics rows projected onto the shortlist and a full-precision SVD — with a mandatory post-hoc residual check at independent kinematics. Full-matrices SVD on $5184+$) columns times out on this box; the two-stage route is both faster and more reliable.

11F. AMP-17: the even-n) relation module identified (new)

11F.1 Result

The even-n) relation count $\kappa_8^{\mathrm{even}}=15$) of §11D.2 is now identified as an explicit $S_4\times S_3$)-module. The reduced 144-class of the S-channel collapses to 72 reversal-classes; relabelings act through $G=S_4\times S_3$) (the odd-slot labels $O\cup\{n-1\}$), four of them, permuted by S_4); the even-slot labels $E\setminus\{1\}$), three of them, by S_3)), and the 15-dimensional relation space is a G)-submodule. Character projection gives a clean **multiplicity-free** decomposition into five distinct irreducibles:

$$
R_{15}\;\cong\;\big[(31)\boxtimes(1^3)\big]\;\oplus\;\big[(211)\boxtimes(21)\big]\;\oplus\;\big[(211)\boxtimes(1^3)\big]\;\oplus\;\big[(1^4)\boxtimes(3)\big]\;\oplus\;\big[(1^4)\boxtimes(21)\big],
$$

with dimensions $3+6+3+1+2=15$). Stable across three independent kinematic seeds; $\chi_R(e)=15$) exactly and every multiplicity is an exact integer.

11F.2 Structural reading

Three honest observations, no formula claimed:

1. **The S_4 (odd-label) leg avoids the top of dominance.** Only (31), (211), (1^4) appear — never the trivial (4) or $(2,2)$. The module is concentrated on the "lower half" of the S_4 lattice, weighted toward the sign rep (1^4) (which carries two of the five constituents).

2. **Not sign-self-conjugate.** Tensoring both legs with their sign reps does *not* fix the constituent set (it maps to a disjoint five-element set involving (4) and $(2,2)$). So the even-n relation space is **not** symmetric under the naive reversal-parity sign twist — a genuine asymmetry, not an artifact of the reversal-even structure of PT_{n-1}.

3. **Contrast with the odd-n fiber modules (AMP-12).** The odd-n signed fiber module was the full regular representation (multiplicity $d_\lambda d_\mu$); here the even-n relation space is multiplicity-free and lands on a specific five-irreducible sliver. The even and odd fiber structures are therefore qualitatively different objects, not two values of one family — consistent with the falsification of the P-S prediction that had implicitly assumed an odd-like reversal structure.

11F.3 Status and the $n=10$ handle

This is an exact, reproducible identification at $m=(4,3)$ (i.e. the $n=9$ S-channel, reduced to $n=8$). It is a single data point in $(m_{\mathrm{odd}},m_{\mathrm{even}})=(4,3)$: a formula for κ^{even} or an even-n fiber law needs the next case, the $n=11$ S-channel (reduced 10-point class, $S_5\times S_4$ acting). The decomposition rule to test there: does the relation module again avoid the dominance-top irreducibles of the odd-label leg and remain multiplicity-free? The AMP-17 machinery (character projection against $\lambda\boxtimes\mu$) ports directly.

11G. AMP-18 (EHF-FACT): does the chamber structure factorize? (new, exploratory)

This section steps from exact combinatorics toward the physics bridge: it asks whether the chamber decomposition is compatible with **factorization**, the property that makes a rational function an amplitude. For the PT basis the native factorization operation is the 2-particle collinear residue (a single PT function has poles only at cyclically-adjacent collinear configurations); the anchor channel $\lambda_j\to\lambda_1$ gives the deletion map $c\mapsto\hat c$ already used in Lemma 2. Status: reproducible, but the **interpretation is open** (see §11G.4).

11G.1 F1 — geometry factorizes (anchor channel), and only there

Each \(n\)-chamber maps under the anchor-channel collinear residue to a **single** \((n-1)\)-chamber: per-channel single-valued at \(n=7\to6\) and \(n=9\to8\) (`amp18_factorization.py`). The discriminating control (`amp18b_control.py`): deleting an **interior** label — not adjacent to the anchor, hence carrying no collinear pole — scatters a source chamber across 9/10 (\(n=7\)) and 34/35 (\(n=9\)) target chambers, essentially maximal. So single-valuedness tracks the pole structure exactly: where there is a physical residue the chambers are respected, where there is none they shred. Caveat: "physical" here means the anchor's own channel (a leg collinear with leg 1, at a fixed position); non-anchor collinear channels are not clean. The chamber grading is anchor-relative.

11G.2 F2 — relations factorize (the content test)

The residue of an identity is an identity: if \(\sum_{c}w(c)\mathrm{PT}_n(c)\equiv0\) then the \(\varepsilon^{-1}\) coefficient gives \(\sum_{\hat c}(\delta_j)_*w(\hat c)\,\mathrm{PT}_{n-1}(\hat c)\equiv0\), *so the collinear residue induces a* ***canonical linear map*** \((\delta_j)_*:R_n(B)\to R_{n-1}(B')\) *on the chamber relation spaces, with branch sign* \(+1\) *(leg right after the anchor) or* \(-1\) *(right before), uniform polar factor dropping and reflection-even* \(\mathrm{PT}_{n-1}\) adding no sign. Verified to \(10^{-20}\) against the direct \(\varepsilon\)-limit (`amp18e`).

The content question is the **rank** of this map: bookkeeping would give rank 0, genuine relation content gives a large rank. Result (`amp18f_f2_clean.py`): the residue images **span the full target relation space** — at \(n=7\to6\) for all 10 chambers (rank \(=r_{B'}=1\), *residual* \(\sim10^{-11}\)); *at* \(n=9\to8\) *for* ***34 of 35*** *chambers (rank* \(=15\)*, residual* \(\sim10^{-13}\)*). So the* \((n-1)\)*-chamber relations are, almost entirely, collinear residues of* \(n\)*-chamber relations: the relation tower is generated by factorization. This is real amplitude content flowing through the map, not position bookkeeping — which is what distinguishes a bridge from a re-coordinatisation.*

Incidental structural fact: every \(n=8\) chamber has size exactly 72 and \(r=15\) (uniform, unlike the varied odd-\(n\) spectrum), and \(15=\kappa_8^{\mathrm{even}}\). The per-chamber \(n=8\) relation space **is** the even module of AMP-17 (§11F).

11G.3 The one gap — RESOLVED as an anchor-artifact (AMP-20)

Exactly one \(n=8\) chamber initially resisted, robustly across three seeds (`amp18h_gap.py`): \(E=\{1,4,5,6\},\,O=\{2,3,7,8\}\), where **anchor-1** collinear residues span only 14 of 15 dimensions (clean 14-decade singular gap). This is **not** an intrinsic obstruction: the missing direction \(v\) is reached once a different reference leg is used. `amp20_gap_dissolve.py` collects anchor-**2** collinear residues (channel \(\langle2,j\rangle\), uniform polar factor) of all \(n=9\) chamber relations as full \(n=8\) relation vectors and finds \(v\in\mathrm{span}(\text{anchor-1}\oplus\text{anchor-2 residues})\) at residual \(1.3\times10^{-15}\). So the 15th relation **is** a factorisation residue, through leg 2's channel rather than leg 1's; the gap was the anchor-1 restriction (cf. cyclic non-covariance, §11G.5), and the earlier "intrinsically even relation" reading is withdrawn. Consistent with the scaffold picture: no single reference leg sees a chamber's relations completely chamber-locally, but the union over anchors reconstructs all of them — the relation content is fully factorisation-generated.

11G.4 What this does and does not establish

Establishes: the chamber relation tower is (almost entirely) generated by anchor-channel collinear factorization — the chambers carry genuine amplitude content compatibly with the one operation that defines amplitudes as physical. Does **not** establish: that this is anchor-independent. Every clean result is in the anchor's own channel; non-anchor channels scatter (§11G.1). Whether anchor-relativity is a *feature* (admissible comparison is intrinsically relative to a reference) or an artifact of the basis is an interpretive question the computation cannot settle. F2 has moved the chamber structure from "possibly bookkeeping" to "carries relation content"; it has not established a cyclically-symmetric, gauge-independent factorisation.

11G.5 AMP-19: the chamber structure is anchor-bound (cyclic-covariance test)

A color-ordered amplitude has full dihedral symmetry (cyclic \(\oplus\) reflection); leg 1 being the anchor is a gauge choice **iff** the chamber decomposition respects that symmetry. We already have reflection covariance (\(\sigma_R\), §11E.2). The cyclic relabeling \(g:i\mapsto i+1 \pmod n\) enacts \(A(1\dots n)=A(2\dots n,1)\); testing whether \(g\) (applied to labels, re-canonicalised) maps chambers to chambers is the cyclic half (`amp19_cyclic_cov.py`).

Result: \(\sigma_R\) maps every chamber to a single chamber (19/19 at \(n=7\), 69/69 at \(n=9\) — positive control passes); the cyclic shift **scatters maximally** (max targets 13/19 and 55/69; only one chamber survives, none under \(g^2\)). So the chamber structure is reflection-covariant but **not cyclic-covariant**: its symmetry group is exactly the dihedral subgroup **fixing the anchor leg** (the reflection through vertex 1, together with the leg-1-fixing relabeling stabiliser \(K_n\)). The chamber decomposition is **anchor-bound** — not a gauge-invariant property of the amplitude, but a reference-fixed scaffold.

Reading. Two honest interpretations. (i) *Artifact:* the chamber tower is a property of the 1-anchored representation, gauge-dependent organising structure, not intrinsic physics. (ii) *Feature:* EHF's primitive is *admissible comparison*, which is intrinsically relative to a reference; a comparison-based substrate should single out one element (the reference = anchor) and be covariant exactly under operations fixing it — which is precisely the observed symmetry group. The cyclic-covariance failure is then the structural signature of a reference, not a defect. The computation cannot decide (i) vs (ii); what it establishes is the operational fact that the chambers are scaffold, not invariant — so any *physical* claim must be cashed in anchor-independent quantities. F2 (§11G.2) is one such instance: an anchor-dependent grading plus anchor-channel factorisation reconstructs the (anchor-independent) relation content — gauge-fixed scaffold computing a gauge-invariant result.

Consequence for §11G.3 (confirmed, AMP-20). The 14/15 factorisation gap was an *anchor-channel* statement; re-anchoring at leg 2 reaches the missing relation (residual 1.3×10^{-15}), dissolving the gap as predicted. No intrinsic obstruction.

11G.6 AMP-21: the chamber spectrum is anchor-invariant

The decomposition is anchor-bound (§11G.5), but the **spectrum** — the multiset of (chamber size, relation count κ) — is identical across all anchors (`amp21_invariance_audit.py`, $n=7$, anchors $1\dots7$): κ-spectrum $[11,7,4,4,4,4,1,1,1,1]$ and size-spectrum fixed). So which cycles form a chamber depends on the reference leg; the statistics of the decomposition do not. This is essentially forced by relabeling-naturality (the cyclic relabeling $g:i\mapsto i+1$ carries the anchor-a decomposition to the anchor-$(a{+}1)$ one bijectively, preserving size and κ), so it is a relabeling-natural invariant rather than a coincidence — but it is a genuine anchor-independent quantity, finer than the standard global dimension counts. It is the candidate for "an invariant the scaffold computes." Whether the κ-spectrum is *new* (not already encoded in the standard KK/BCJ/color-kinematics toolkit) is a literature-placement question, addressed in §11H.

11H. Literature placement (honest novelty audit)

Before any novelty claim, a placement against the amplitudes literature (searches June 2026). The result is substantially deflationary and should be read as such.

What is standard (not ours).
- The objects. The canonical Hamilton cycles are the $(n-1)!/2$ cells ("chambers") of the real moduli space $\mathcal M_{0,n}(\mathbb R)$, *each carrying a Parke-Taylor form; this is textbook moduli-space/CHY material (e.g. arXiv:1906.02099).* ***Terminology clash:*** *in the literature "chamber" = one ordering = one Hamilton cycle; our "chamber" = a Π-fiber = a *grouping* of these by parity class. Different object, same word.*

- The relation space. The $(n-2)!$-dimensional KK basis and the $(n-3)!$ BCJ basis are standard (Kleiss–Kuijf; BCJ; e.g. arXiv:1708.08701, 1812.01168). Our recovery of KK rank $(n-2)!$ is a rederivation.
- The representation theory. Decomposing the color-ordered relation space into irreducibles of S_n is an established programme (Edison–Naculich, arXiv:1207.5511), with free Lie algebras / the Lie operad central. Our "KK span $\cong$ Lie$(n-1)$" is Ree's theorem, already flagged as anchor not novelty.
- Factorisation. Collinear residues of Parke–Taylor forms on moduli boundaries are bread-and-butter; the F2 "relation tower generated by collinear residues" is most likely a rediscovery of known boundary structure, valuable as confirmation but not new.

What did not surface in the search (candidate-new, unverified).
The specific Π-fiber ("chamber") grouping of the moduli chambers by parity class, and its anchor-invariant κ-spectrum (§11G.6). This is a *different* grading from the S_n-irrep decomposition. Whether it carries information **beyond** the known S_n-irrep content is the decisive open question and is **not yet settled**: if the κ-spectrum is a function of the S_n-irrep multiplicities, it is a repackaging of Edison–Naculich; if it is a genuinely different (non-S_n-equivariant) invariant, it is new.

Decisive computation (AMP-22 / AMP-22b, done).
- *Subspace test (`amp22_decisive.py`).* The chamber-local relation subspace R^{cl} (span of all relations supported within a single Π-fiber) is **maximally anchor-dependent**: $R^{\mathrm{cl}}(\text{anchor }1)$ and $R^{\mathrm{cl}}(\text{anchor }2)$ are both 38-dim at $n=7$ (1-dim at $n=5$) with **trivial intersection** and principal angles up to 90° — the two anchorings give essentially disjoint subspaces. An S_n-isotypic component is anchor-independent by construction; this is the opposite, so the chamber-local structure is not a canonical S_n object.
- *The known invariant (`amp22b_sn_decomp.py`, Murnaghan–Nakayama table validated orthonormal).* $R_5=(3,1,1)$; R_7 (dim 240) $=2\,(5,1,1)\oplus(4,3)\oplus(4,2,1)\oplus(4,1,1,1)\oplus2\,(3,3,1)\oplus2\,(3,2,1,1)\oplus(2,2,2,1)\oplus(2,2,1,1,1)\oplus(1^7)$. The chamber-local relations are a small (38/240), anchor-dependent slice of this; the κ-values $\{11,7,4,1\}$ are **not** S_7-irrep dimensions ($\{1,6,14,15,20,21,35,\dots\}$).

Verdict. The chamber κ-spectrum is **not** a repackaging of the S_n-isotypic decomposition: it is the dimension-spectrum of a maximally anchor-dependent (non-equivariant) subspace, with values that are not isotypic dimensions. This is the first quantity in the programme that survives the novelty audit as genuinely distinct from the standard amplitude/representation-theory toolkit — a candidate new invariant.
Residual caveat — now CLOSED (AMP-23, `amp23_branching.py`). The remaining worry was that κ might be computable from the S_n-multiplicities via a Mackey/branching formula. It cannot: every chamber is a $3|3$ parity split of $\{2,\dots,7\}$, so all chambers have **conjugate** Young subgroups $S_1\times S_3\times S_3$. A Mackey/branching formula depends only on the conjugacy class of the subgroup, hence would assign a **constant** κ to all chambers; but κ takes values $\{1,4,7,11\}$. Explicit witness: chambers with even-non-anchor sets $\{5,6,7\}$ ($\kappa=11$) and $\{4,6,7\}$ ($\kappa=7$) are related by $g=(4\,5)\in S_6$ fixing leg 1 — conjugate subgroups, different κ. The branching module $\mathrm{Res}^{S_7}_{S_1\times S_3\times S_3}(R_7)$ *is one fixed $S_3\times S_3$-module (dim 240, computed and dimension-checked), identical for every chamber, while the chamber κ's differ. So κ genuinely depends on how each split sits relative to the reflection axis through leg 1 — non-equivariant data not in $\{m_\lambda\}$.*

Status of the candidate invariant. It now survives: literature placement (did not deflate), anchor-invariance (AMP-21), transversality across anchors (AMP-22), distinctness from S_n-isotypic dimensions (AMP-22b), and non-derivability from S_n-branching (AMP-23).
Remaining before any external novelty claim: $n=9$ confirmation of the same non-equivariance, and a targeted literature search on this specific parity-fiber object. No external claim until then.

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12. Pre-Registered Predictions (open)

1. *(Confirmed, AMP-10.)* $(10,11)$-paired $n=11$ chambers: all **70** split fiber pairs have size gap exactly 576; the state-level Lemma-1 injection (signs, $|X|=576$, pattern $c_1=10$, $c_{10}=11$) *verified exactly on tested pairs; on the first relation-bearing pair (sizes 3456/4032), $r_A=r_B=6$ at 12.5/12.4-decade gaps, with Lemma-2 vanishing and both relation-space inclusions at 10^{-14}. The sibling mechanism is now a three-n law. Numerics note: the initial run produced an artifactual inequality with a 0.0-decade gap; the failure was self-diagnosing, fixed by row/column equilibration, and the fixed solver was validated against the known $n=9$ ground truth ($r=177$), 14.1 decades) before re-running.*

2. *(Resolved — PROVED, AMP-16; §11E.)* $K_{11}\supseteq\langle(2,3),(10,11),\sigma_R\rangle$. σ_R is a theorem for all n (canonical-flip argument); $(10,11)$ follows from Lemma 2; $(2,3)$ from the mirror Lemma 2$^\prime$, with mod-p determinant certificates at $n=9$ and $n=11$. Decisive witness chamber C605, and a 16/16 small-label vanishing sweep at $n=9$; splitting controls fail. No open pre-registered predictions remain.
3. *(Decided.)* AMP-09 falsified both registered hypotheses of §11.4 and measured the law $\mathrm{rank}(V|_C)=\min(342,|C|-r_C)$; *the boundary-anomaly sub-prediction also failed. New open constant:* $\mathrm{cap}(7)=20$, $\mathrm{cap}(9)=342$.

Bookkeeping caveat: at each n there are 20-type split fiber pairs but 19 sibling pairs; the remaining pair involves the unique unrealised fiber and is degenerate for Lemma 1 as stated.

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13. Theorem-Style Results Currently Supported

1. **Identification.** The odd RAF state space is the trace basis modulo reflection; odd-n amplitudes are reflection-odd sections over it.
2. **Fiber connectivity (candidate).** Chambers = Π-fibers, exact at $n=7,9,11$.
3. **Flip-complement pairing (exact).** $|C(\Pi)|+|C(\mathrm{flip}\,\Pi)|=((n-1)/2)!^2$; one unrealised fiber per n.
4. **Statistical visibility.** Π-membership explains 31.5%/22.0% of $\log|PT|$ variance at $n=7/9$ at overwhelming significance.
5. **Negatives.** No filter amplitude-smoothness; no KK block-diagonalisation; representation type does not determine r; no parent-(size,\,r) recursion.
6. **Chamber-local relations.** 38 at $n=7$; 1962 at $n=9$; σ_R-pairs equal; only the central 144-chamber C carries relations among the 144s.
7. **Explicit identities.** All-±1 basis in the 36-chamber; $\mathbb{Z}_3$ *orbit-difference structure; one identity exact.*
8. ***Boundary stabiliser law (now proved, AMP-16).*** *$K_7\cong D_4$ exactly; and $K_n\supseteq\langle(2,3),(n-1;n),\sigma_R\rangle$ is a THEOREM for $n=7,9,11$: σ_R by a canonical-flip argument (all n)), $(n-1;n)$ by Lemma 2, $(2,3)$ by the mirror Lemma 2$^\prime$ (mod-p determinant certificates). The $n=11$ case was a confirmed pre-registered prediction (§12.2) before proof.*
9. ***Sibling equality: unconditional theorem (AMP-15).*** *Lemma 1 proved; Lemma 2 PROVED at $n=9$ and $n=11$ (single-collinear residue argument plus exact mod-p determinant certificates, §11D); hence $r_A=r_B$ for all 19 pairs at $7\to9$ and all split pairs at $9\to11$; ladder gaps 36 and 576 re-derived. The $n=11$ confirmation (70/70 size gaps; $r_A=r_B=6$ on the first relation-bearing pair) preceded the proof as a pre-registered prediction.*
10. ***BCJ chamber law.*** *The BCJ-reduced span is non-generically chamber-aligned (random-subset control 100/100 null); rb is size-uniform for $|C|\ge20$ with $rb=|C|-\mathrm{cap}(n)$; the PT-level 4/4/1 irregularity equalises at the BCJ level.*

11. **Fiber law (new).** *Flip-pair fiber-unions are relabeling-covariant, forcing uniform fiber invariants. PT fiber relation count κ: 11 / 177 (uniform across all tested pairs, gaps 14–15 decades), with cross-chamber relations binding complementary chambers; BCJ fiber rank = cap: 20 / 342 (confirmed on $n=9$) unions against generic values); large-chamber kernels equal the fiber kernel.*
12. **Fiber modules (new).** *The full signed fiber module is the regular representation of $S_m\times S_m$); the relation/span multiplicity tables are computed exactly at $n=7,9$) and are uniform across fibers; $U\cong\mathrm{Lie}(n-1)$) (verified, all classes); multiplicity data provably cannot determine κ).*
13. **Explicit cross-chamber identities (new).** *On the $(32,4)$) fiber pair, exact rank 25 over $\mathbb{Q}$); the relation lattice surjects onto the tiny-chamber coordinates with unit pivots; every tiny-chamber amplitude is an all-±1) combination of complementary-chamber amplitudes (supports 6,6,6,10), verified exactly; the four identities form one stabiliser orbit.*

9a. **Small-label vanishing lemma (Lemma 2$^\prime$), new, AMP-16).** *For chambers with $2\in O,3\in E$), every chamber relation vanishes on the corner $\{c_1=2,c_{n-1}=3\}$); proved by a single-collinear residue argument plus exact mod-p) determinant certificates at $n=9,11$); the engine behind $(2\,3)\in K_n$). Confirmed 16/16 across relation-bearing B-form chambers at $n=9$).*

14a. **S-channel structure (new, AMP-15).** Pre-registered P-S antisymmetry falsified consistently: the even-n) reduced 144-class has rank 57, giving the first even-n) fiber relation count $\kappa_8^{\mathrm{even}}=15$); chamber relations do not vanish on Y_2); the surviving law constrains symmetrised S)-weights to the 15-dimensional even-fiber space.

14b. **Even-n) relation module identified (new, AMP-17).** The 15-dimensional even-fiber relation space is the multiplicity-free $S_4\times S_3$)-module $(31)\boxtimes(1^3)\oplus(211)\boxtimes(21)\oplus(211)\boxtimes(1^3)\oplus(1^4)\boxtimes(3)\oplus(1^4)\boxtimes(21)$); concentrated on the lower-dominance S_4)-irreducibles, not sign-self-conjugate, and qualitatively unlike the regular-rep odd-n) fiber modules of AMP-12.

14. **The distinguished covector (new).** The $(28,8)$) pairs are the unique cokernel-1 cases; the canonical covector φ) is the all-$+1$) sum over the four states with the smallest label at the interior position, exact and unique over $\mathbb{Q}$), stabiliser-trivial, and **not** in the BCJ evaluation span at generic kinematics.

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14. Open Problems

1. Prove fiber connectivity for all odd $n\ge7$) (known $n\le11$)).

2. A uniform all-n proof of the reduced-family independence behind Lemma 2 (currently one exact determinant certificate per n); and the even-n fiber law: the $\kappa_8^{\mathrm{even}}=15$ module is now identified (§11F), so the open step is the next case — the $n=11$ S-channel under $S_5\times S_4$ — to test whether the multiplicity-free, dominance-lower pattern persists and yields a κ^{even} formula.
3. *(Done, AMP-16.)* Both $n=11$ predictions resolved: §12.1 confirmed, §12.2 confirmed and proved.
4. Derive the closed form of the fiber multiplicity tables (§11B.2). Multiplicity data alone is provably insufficient (§11B.4); the formula must come from the structure of the Parke-Taylor evaluation map, e.g. in Specht bases.
5. Prove the BCJ cap law $rb_C=\max(r_C,|C|-\mathrm{cap}(n))$ and explain the isolated $n=7$ size-20 anomaly (which does NOT recur at the $n=9$ boundary).
6. The equality direction of the boundary stabiliser: the $\supseteq$ inclusion is now a theorem (§11E), and $K_7=D_4$ is exact; prove $K_n=\langle(2\,3),(n-1\;n),\sigma_R\rangle$ (i.e. no further stabilising relabelings) for $n\ge9$, giving $K_n\cong D_4$ for all odd $n\ge7$.
7. Derive the splitting of κ into $r_C + r_{C_\mathrm{flip}} + \mathrm{CROSS}$ (the chamber-level quantities are the diagonal part of the uniform fiber structure); characterise the cross-chamber relations explicitly.
8. Literature placement of the explicit identities and of the bi-adjoint chamber alignment before novelty claims.
9. Explain the relation-count size thresholds and the uniqueness of the central 144-chamber.

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15. Conservative Statement Without EHF Language

> Identify canonical Hamiltonian cycles on K_n with the trace basis of n-point colour-ordered tree amplitudes modulo reflection. For odd n, the parity-position invariant Π partitions this basis into sectors coinciding with the preserve-world connected components at $n=7,9,11$, with the exact size pairing $|C(\Pi)|+|C(\mathrm{flip}\,\Pi)|=((n-1)/2)!^2$. The annihilator of the Kleiss-Kuijf subspace contains sector-supported relation spaces of total dimension 38 at $n=7$ and 1962 at $n=9$, invariant under the boundary group generated by the transpositions of the two smallest and two largest labels and label reversal — confirmed at $n=9$ as a pre-registered prediction. Along the exact core-growth ancestry, sibling sectors carry equal relation dimensions, following from a proved injection lemma for the adjacent transposition together with a vanishing lemma proved by a collinear residue argument and exact determinant certificates. At the kinematics-dependent level, the BCJ-reduced span (column space of the bi-adjoint double-partial matrix) is non-generically aligned with the sectors: sector-restricted ranks carry exact, kinematics-independent deficits absent in 100/100 matched random controls, with sector-local BCJ relation counts uniform within size classes and equal to $|C|-20$ for $|C|\ge24$ at $n=7$.

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16. EHF Relevance

The AMP series is the first exact, non-statistical contact between RAF structures and an externally defined physical formalism, now extending to the kinematics-dependent layer that the KLT kernel squares into gravity amplitudes. It does **not** establish that RAF derives gauge theory or gravity, that chamber dynamics corresponds to a physical process, or novelty of individual identities prior to literature placement. The framework makes no contact with gauge-group selection, matter content, or chirality; Standard-Model and GUT identifications are out of scope.

The supported statement:

> Structures discovered purely from local admissible comparison exactly grade and organise relation structure of tree-level gauge amplitudes at both the constant-coefficient (KK) and momentum-kernel (BCJ) levels, including two laws established or supported by pre-registered prediction (the boundary stabiliser; the sibling mechanism) and one newly discovered size-uniform BCJ chamber law with registered competing explanations awaiting the $n=9$ decision. The amplitude/double-copy direction is the physics bridge of the programme.

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17. Reproducibility

Standalone scripts (independent implementations; fixed seeds):

- `replicate_opw.py`; `opw_fiber_connectivity.py` (fiber connectivity, $n=9$ validation and $n=11$).
- `amp01_probe.py`; `amp02_kk_decomposition.py`, `amp02_verify.py`; `amp03_extract_relation.py`.
- `amp04b_isotypic.py`, `amp04c_global_s6.py`, `amp04d_partition_stabilizer.py`, `amp04e_true_stabilizer.py`.
- `amp05_n9_spectrum.py`; `amp06_r_ladder.py`; `amp07_sibling_mechanism.py`.
- `amp08_bcj_kernel.py` (bi-adjoint construction; sign-convention selection at $n=5,6$); `amp08_n7_main.py` ($n=7$ chamber ranks, three kinematic points); `amp08_controls.py` (matched random-subset control; common-vanishing-subspace test).
- `amp09_n9_bcj.py`, `amp09b_large.py`, `amp09c_boundary.py` ($n=9$ per-chamber bi-adjoint: validation, H1/H2 decision, boundary test).
- `amp10_n11_sibling.py`, `amp10b_relation_pair.py`, `amp10c_fixed.py` ($n=11$ sibling-mechanism prediction: universal size-gap check; relation-pair scan; equilibrated solver with $n=9$ ground-truth validation).
- `amp11_kernel_geometry_n7.py`, `amp11b_fiber_law.py` (kernel geometry; the fiber law at PT and BCJ levels, $n=7$ and $n=9$).
- `amp12_fiber_modules.py`, `amp12b_span_side.py`, `amp12c_lie_check.py` (fiber-module decompositions and multiplicity tables; full-module and span-side ledger; Lie-module identification of the KK span).

- `amp13_cross_extraction.py` (exact extraction of the four cross-chamber identities on the (32,4) fiber pair; onto-projection and orbit checks).
- `amp14_cokernel.py` (the distinguished covector of the (28,8) pair: exact extraction, uniqueness, stabiliser character, corrected BCJ-visibility check, cokernel table).
- `lemma2_proof.py` (AMP-15: proof certificates for the vanishing lemma at $n=9$ and $n=11$; S-channel rank measurements; falsification tests on all relation-bearing B-form chambers).
- `amp16b_k11_split_test.py`, `amp16e_k11_decisive.py`, `amp16f_small_corner_sweep.py` (AMP-16: the splitting stabiliser test with the transported chamber C605 and splitting controls; the two-stage decisive $(2\,3)$ corner test validated against the $r=6$ ground truth; the 16/16 small-label vanishing sweep at $n=9$).
- `lemma2prime_cert.py` (AMP-16: mod-p determinant certificates for the small-label vanishing lemma at $n=9$ and $n=11$).
- `amp23_branching.py` (AMP-23: closes the branching caveat -- kappa varies across conjugate chambers, so it is not Mackey/branching-determined; explicit witness and the fixed Res to S3xS3).
- `amp22_decisive.py`, `amp22b_sn_decomp.py` (AMP-22: chamber-local subspace is maximally anchor-dependent across anchors; validated S_n-irrep decomposition of R_n; the kappa-spectrum is transverse to the isotypic data -- candidate new invariant).
- `amp21_invariance_audit.py` (AMP-21: anchor-invariance audit; the (size, kappa) chamber spectrum is invariant across all anchors at n=7).
- `amp20_gap_dissolve.py` (AMP-20: dissolves the 14/15 gap — the missing relation is an anchor-2 factorisation residue, residual 1e-15).
- `amp19_cyclic_cov.py` (AMP-19 / EHF-FACT-03: cyclic-covariance test; chamber structure is reflection-covariant but anchor-bound, symmetry group = dihedral subgroup fixing the anchor).
- `amp18_factorization.py`, `amp18b_control.py`, `amp18f_f2_clean.py`, `amp18h_gap.py` (AMP-18 / EHF-FACT: F1 chamber-geometry factorisation and the pole-vs-interior control; F2 residue map on relation spaces, near-surjectivity at n=7,9; the one robust 1-dim gap and its identification).
- `amp17_even_module.py`, `amp17b_verify.py` (AMP-17: $S_4\times S_3$ character-projection decomposition of the 15-dimensional even-n relation module; three-seed robustness and the sign-twist structural checks).

Result files: `amp01_results.json` ... `amp06_ladder.json`, `opw_replication_results.json`.